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Cue-Based Approach for User Activity Recognition in Multi-User Virtual Reality and Mixed Reality: A Scoping Review

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Abstract

Multi-user virtual and mixed reality systems require understanding how users behave, interact, and collaborate within these spaces. Single-user approaches focus on recognizing an individual's actions in isolation. However, multi-user scenarios require multiple participants to understand and react to each other's actions in a shared environment. In this paper, a scoping review was conducted to identify cue-based approaches for user activity recognition in multi-user VR and MR systems. The scoping review included 20 studies and explores how cues are developed and integrated in systems, their effect on user experience, and what opportunities and challenges exist in design and integration of cue-based approaches for user activity recognition.

Index Terms: Cues, Mixed Reality, Multi-User, User Activity Recognition, Virtual Reality

1 Introduction

Multi-user virtual reality (VR) and mixed reality (MR) systems have the potential to support remote or co-located users to work together in a shared immersive environment, interacting with virtual and real objects as if they were physically co-present [32]. This shared space can result in a strong sense of co-presence, the feeling of "being together" with others [34]. Co-presence depends on how well each user can perceive the presence and intentions of their collaborators [34].

A key factor underpinning co-presence is awareness of others' activities [19]. In human-computer interaction research, awareness is defined as an understanding of the activities of others that provides context for one's own actions [7]. In a physical shared workspace, collaborators naturally maintain high mutual awareness through sight, sound, and peripheral cues, for example, by noticing a partner's movement out of the corner of one's eye [16]. Moll et al. [16] found that people working co-located have better awareness of the presence of others and their activities than those working remotely, a condition that immersive systems must deliberately recreate.

This challenge is especially acute in task-oriented scenarios. When users manipulate shared objects, for example, handing a virtual tool to a teammate, they must know where the other user is focusing and what they plan to do [20]. Kim et al. noted similar observations that users should be aware of the movement of themselves and others in VR collaborative tasks to coordinate their actions to move objects in the shared space [13]. Lacking this awareness can cause repeated actions or even collisions in a virtual task space.

The design space for cue-based approaches is fragmented with studies spanning different cue types [22, 32], tasks [26], and co-located versus remote settings [1, 17]. To synthesize work on cue-based approaches for user activity recognition, this scoping review aims to answer the following research questions:

RQ1: What cues have been developed for user activity recognition and how do they influence the user experience in multi-user VR and MR systems?
RQ2: What are the opportunities and challenges in designing and integrating cues for user activity recognition in multi-user VR and MR systems?

This paper conducts a scoping review focused on cue-based approaches for user activity recognition in multi-user VR and MR to examine how cues are used to help users understand each other's actions in a shared environment, and what evidence exists regarding their effectiveness. The main aim of this review is to provide an overview of current cue-based approaches and identify challenges and opportunities for designing cues for user activity recognition in multi-user VR and MR systems.

2 Related Works

Several recent surveys and literature reviews have examined aspects of multi-user interaction in extended reality (XR) applications that are discussed below in the related works followed by studies on cue-based approaches for user activity recognition in XR systems.

2.1 Systematic Reviews on Multi-User XR

de Belen et al. [3] performed a systematic review of collaborative MR systems and identified key factors that support effective collaboration. They emphasized annotation techniques as important communication cues that help collaborators understand each other's actions. In their synthesis, annotation cues complement other approaches for cooperative object manipulation techniques and user perception studies to reduce cognitive load during task understanding and completion.

More recently, Assaf et al. [2] surveyed XR collaborative systems with a focus on workspace awareness and visual cues. They analyzed a broad range of XR prototypes and classified the kinds of shared awareness problems and visual cues that researchers have employed. Another systematic review by Wang et al. [29] looked at XR mediated multi-user social engagement. It highlights verbal and non-verbal cues as important cues for engagement in multi-user social VR to convey collaborators' information and intentions.

Sayadi et al. [24] reviewed 20 years of social VR literature and found that non-verbal and spatial cues can improve teamwork by improving individual users' sense of presence, team dynamics, and enabling novel interaction techniques. These broader surveys showed a consensus that cues are important for VR collaboration, but they largely cataloged the phenomena rather than evaluating specific cue designs.

2.2 Cues for User Activity Recognition

Some individual studies have also examined cue-based techniques in specific VR and MR contexts. Piumsomboon et al. [20] con-

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ducted an experiment to explore effect of awareness cues in a mixed AR–VR collaboration task. The participants worked in pairs, one in AR and one in VR, on a task for object finding and placement. The authors concluded that even simple visual overlays of a collaborator's attention can support shared understanding.

Moll et al. [16] studied collaborative object building in a 3D environment, pairing a sighted and a blindfolded user. They added spatialized sound cues to signal each partner's actions. The participants were able to remain aware of ongoing activities and receive feedback on their actions, demonstrating the effectiveness of sound cues in maintaining situational awareness.

Kim et al. [13] investigated multimodal awareness in VR collaboration. They presented a dual awareness framework that provides distinct feedback for selves and for others. Participants received visual cues and haptic feedback to differentiate their actions from those of their collaborators. During a virtual object handover task, participants using dual-awareness cues completed tasks more quickly and reported higher situational awareness. These findings suggest that users benefit from understanding both their own and others' movements.

3 Method

A scoping review protocol was created based on PRISMA guidelines[1] for conducting the scoping review. The title, abstract, and keyword search was made in three databases: Scopus, Web of Science, and IEEE Xplore, filtered for publication years 2015-2025 and English language. The search string used in the databases was as follows:

("user activity" OR "user behavior" OR "behavior analysis" OR "user interaction" OR "social interaction" OR "social engagement" OR "communication") AND ("attent*" OR "recogni*" OR "track*" OR "regist*") AND ("cue*") AND ("collaborat*" OR "cooperat*" OR "interact*" OR "co-locate*" OR "remote*" OR "multi-user" OR "social" OR "share*") AND ("extended reality" OR "virtual reality" OR "mixed reality" OR "XR" OR "VR" OR "MR" OR "immers*")

The ACM Digital Library was not included as a database source because the search string returned no records there. Iterative modification of cue-related terms were made such as replacing it with "cue", "cues", and removing the term completely which also yielded no results. However, most ACM Digital Library publications are indexed in Scopus and relevant ACM records were retrieved through Scopus and included in the review via that database.

The first parentheses contains synonyms for user activity followed by synonyms for recognition. The terms are connected with the term cue since the focus of the scoping review is cue-based approaches for user activity recognition. The terms related to multi-user and extended reality (XR) technologies are then included to cover multi-user XR applications.

The search results for the above search string in the three databases are shown below in Table 1. A total of 396 documents were retrieved from the databases. After removing 131 duplicates, 296 papers remained for the first round of title and abstract screening.

Database	Number of Results
Scopus	206
Web of Science	151
IEEE Xplore	38

Table 1: Number of results for the searches made in three databases.

Two reviewers independently carried out title and abstract screening marking papers as included, excluded, or maybe when in doubt about their decision along with short notes to justify their decisions. The collected papers were screened using the following inclusion and exclusion criteria shown in Table 2. The reviewers had a 99% agreement rate and any conflicts in decisions were resolved after discussion.

[1] https://www.prisma-statement.org/

Inclusion Criteria	Exclusion Criteria
VR or MR applications.	Augmented reality (AR), 360-degree videos, and asymmetric applications that do not use at least VR or MR as one of the systems.
Multi-user scenarios.	Single user scenarios.
Studies that include design or implementation of cues.	Studies that do not explain how user activity is recognized or conveyed to other users.
Systems with human to human interaction.	Systems with human to agent, human to robot, or agent to agent interactions.
Cues designed for user activity recognition.	Cues designed for emotion recognition.

Table 2: Inclusion and exclusion criteria for paper screening.

After title and abstract screening, 34 papers were selected for full-text screening. Out of these, 20 papers were included in the review following the same screening process. An overview of the complete screening process can be seen below in Figure 1.

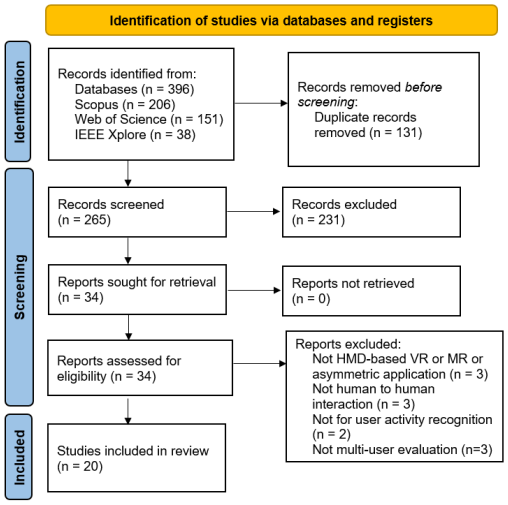


Figure 1: PRISMA flowchart to summarize the screening process for the scoping review.

A data extraction form was developed for data extraction and synthesis. The data extracted from the papers included the title, author names, country of the study, and publication year. The data extracted to answer research questions were XR application type, i.e., if it is a VR or MR application; application area such as gaming, meetings, etc.; the cues developed and type of cue; what aspect of user experience was evaluated; design and implementation of the cues; and evaluation design. A thematic analysis was conducted to synthesize the findings from the extracted data.

4 Overview of Included Studies

In this section, we present a summary of the key characteristics of the studies included in this scoping review. We report the authors, publication year, and country of each study to provide a descriptive overview of the literature on cue-based approaches for user activity

recognition in VR and MR applications.

Of the 20 included studies published from 2015 to 2025, 55% were published in the last three years, with the highest number of publications occurring in 2024 (n=5). This trend reflects a growing interest in cue-based methods for activity recognition in multi-user VR and MR applications.

The studies were conducted across ten countries and predominantly in United States of America (USA), United Kingdom (UK), Australia, and Germany with fewer studies originating from China, Japan, India, Canada, Spain and Singapore.

5 XR Application Type and Area

This section synthesizes the types of XR applications developed and the corresponding application areas addressed by the included studies. We distinguish between VR and MR systems and describe the domains and tasks for which cue-based user activity recognition approaches were developed.

5.1 XR Application Types

The majority of the reviewed systems were implemented in VR environments (n=17). VR-based applications were predominantly used to investigate user activity recognition in fully immersive, simulated settings. In contrast, a smaller subset of studies employed MR (n=3), where virtual elements were overlaid onto the physical environment to support task-oriented and collaborative activities.

VR applications were commonly used to study both individual and social interactions [5, 6, 8, 10, 11, 14, 15, 28, 30, 31] using controlled virtual environments to observe user behavior in varied scenarios. MR applications, while fewer in number, focused primarily on collaborative [33] and spatial tasks [18], highlighting the relevance of real-world context in user activity recognition in multi-user scenarios.

5.2 XR Application Areas

A wide range of application areas were identified for VR and MR systems developed in identified studies. These can be broadly categorized into social interaction, collaborative work, task-oriented problem solving, and training or professional scenarios.

Majority of VR-based studies focused on social interaction scenarios (n=9), including social conversations [5, 10, 28, 30], meetings [14, 15], and interactions within virtual rooms [8], outdoor spaces [31], or VR events [6]. These studies examined user behavior in individual, pair-based, and group settings, reflecting the growing interest in understanding social dynamics in immersive virtual environments.

Several VR applications targeted task-oriented activities (n=6), such as puzzle solving [9], drawing tasks [35], data visualization [4], urban planning [27], assembly tasks [23], and intruder detection within game-like scenarios [21]. These tasks were conducted by individuals, pairs, or small groups and often required coordinated actions, decision-making, or problem-solving, making them suitable for cue-based activity recognition.

A smaller number of VR studies addressed training-oriented applications, including medical training [12] (n=1) and training for picking up social cues for children with autism [11] (n=1), where accurate recognition of user actions is critical for performance assessment and feedback.

MR applications were primarily associated with collaborative and spatial tasks, such as collaborative assembly [33] (n=1), social interaction during gameplay [25] (n=1), and spatial planning tasks [18] (n=1) related to architecture or memory-based object manipulation. These applications involved pair-based interactions and integrated virtual cues within the physical environment to support activity recognition.

6 Cue-Based Approaches

In the reviewed studies, a diverse range of cues have been developed to register and convey user behavior and interaction patterns. These cues can be broadly categorized by the type of sensory modality they use which are visual, auditory, and tactile.

Visual cues are the most prevalent across the studied systems. These include cues derived from facial expressions, gaze direction, pointing, gestures, head movements, and other body movements [11, 25, 27, 35]. For example, gaze and hand gestures are commonly used both for social interaction recognition and attention guidance [5, 9]. Non-verbal cues such as posture and spatial positioning are also frequently captured visually for identifying social dynamics [14] and intruder awareness in shared virtual spaces [21].

Auditory cues complement visual cues in several systems. These include speech-based cues, verbal instructions, and non-verbal audio signals such as gestures with sound or voice tone used to detect social interactions and guide attention [4, 12, 28, 31]. Tactile cues appear less frequently but are significant in systems aiming to simulate multi-sensory interactions. Tactile feedback is primarily used in conjunction with social or non-verbal cues, supporting interactions that involve touch or haptic signaling between users [10, 15, 30].

Integration of multisensory cues is a recurring theme in these approaches. Many systems combine visual and auditory cues, e.g., gaze with verbal cues [4, 28], or visual and tactile cues, e.g., social cues with haptic feedback [10, 15, 30]. Some systems also incorporate physiological cues, such as heart rate or other biometric signals, alongside visual observations to infer emotional or attentional states [23].

Table 3 below summarizes authors, year of publication, country where the research was conducted, XR application type and area, and cue types developed for studies included in the review. The country assignment was determined based on the authors' institutional affiliations and/or the reported location of data collection, where available.

6.1 Design and Implementation

Gaze and Visual Attention Cues: Teo et al. [27] detected facial expressions of users which were then converted to emojis to share emotions. Emojis were animated using Unity's particle system where they appear and float away from a designated position. They were then displayed either on the hands of users or objects when the user or their partner performed an action. Caruana et al. [5] developed two tasks using Unity in a virtual environment with avatars to represent users. The non-verbal cues gaze and pointing were tracked using sensors and eye tracker. Ichino et al. [8] selected three styles of visualization for gaze cue arrow, bubble, and mini avatar that were visible while gaze was active and were deactivated after the calculated gaze dwell time for one-sided and joint gaze. Bovo et al. [4] developed a VR scene with participants positioned in front of four panels that had data visualizations from three databases. They converted speech to text using Google speech-to-text service in real time and visualized gaze as a cone of vision for conveying where the one of the user was looking. Jing et al. [9] developed a prototype that combined gaze, hand pointing, gesturing, and heart rate visualizations for users to freely exchange bi-directional real time cues. Three gaze behaviors were visualized for browse state when there is active eye movements, focus statements when dwell time is over 500ms, and joint state when collaborators' gaze overlap for over 200ms. A virtual ray was casted when users point using fingers and hands were tracked and visualized for displaying gestures. Vinnvikov et al. [28] developed a virtual environment with a virtual microphone to represent the listener. The scene included animation for avatars as they spoke to mimic lip and facial movements which were then displayed to the user. A real-time gaze-contingent system was assembled to work with the

Authors	Year	Country	XR Type	Application Domain	Cues Developed	Types of Cues
Teo et al. [27]	2025	Australia	VR	Urban Planning	Emotional cues	Visual
Zhang et al. [35]	2025	UK	VR	Drawing	Social cues	Visual, Tactile
Jung et al. [10]	2024	USA	VR	Social Interaction	Non-verbal cues	Auditory, Tactile
Caruana et al. [5]	2024	Australia	VR	Social Interaction	Non-verbal cues	Visual
Qin et al. [21]	2024	China	VR	Gameplay	Awareness cues	Visual
Ichino et al. [8]	2023	Japan	VR	Social Interaction	Gaze	Visual
Lee et al. [15]	2024	USA	VR	Meetings	Attention guidance cues	Visual, Auditory
Yan et al. [33]	2023	China	MR	Assembly Task	Non-verbal, Spatial cues	Visual
Bovo et al. [4]	2023	UK	VR	Data Visualization	Gaze, Verbal cues	Visual, Auditory
Jing et al. [9]	2022	Australia	VR	Puzzle Solving	Non-verbal cues	Visual
Williamson et al. [31]	2022	UK	VR	Social Interaction	Social cues	Visual, Auditory
Kurzweg et al. [14]	2022	Germany	VR	Meetings	Communication cues	Visual
Wieland et al. [30]	2022	Germany	VR	Social Interaction	Non-verbal cues	Auditory, Tactile
Sayis et al. [25]	2020	Spain	MR	Gameplay	Body cues	Visual
Jyoti et al. [11]	2019	India	VR	Training	Non-verbal cues	Visual
Vinnikov et al. [28]	2017	Canada	VR	Social Interaction	Non-verbal cues	Auditory
Müller et al. [18]	2016	Germany	MR	Spatial Planning	Spatial cues	Visual
Kim et al. [12]	2016	USA	VR	Training	Non-verbal cues	Visual, Auditory
Sasikumar et al. [23]	2024	Singapore	VR	Assembly Task	Physiological cues	Visual
Chang et al. [6]	2025	USA	VR	Social Interaction	Non-verbal cues	Visual

Table 3: Summary of authors, year of publication, country where the research was conducted, XR application type and area, and cue types developed for the studies included in the scoping review.

virtual environment. Jyoti et al. [11] designed a system with three modules for providing joint attention cues to users using a virtual environment with virtual objects and avatars. The avatar initiated eye contact with the user and then presented gaze-based, head turn, finger-pointing cues. Real-time physiological data was collected using Biopac MP150 and stored in a database synchronized with task progression using event-related markers.

Full Body Movement and Gestures for Non-Verbal Cues: Zhang et al. [35] used full-body inverse kinematics (IK) tracking for identifying movements such as pointing, miming or shifting posture and social touch interactions like hand shake and high five and full body tracking for hugs. Players used hand and body movements to convey these non-verbal cues during multiplayer social interaction. Yan et al. [33] developed multiple components of a system in which experts could use gestures to control the degree of information displayed, perceive expert's gaze and gesture information, and enable the users to share 3D virtual replica copy spatial cues across local MR and remote VR sides. Sayis et al. [25] developed a full-body interaction environment, an installation-based virtual environment. The visual interface was configured as circular projection on the floor generated by images from projectors and the interactive experience had visual and auditory components. Two players navigated the virtual space together and used verbal communication, facial expressions, and body language to interact. Williamson et al. [31] developed a virtual environment for outdoor meetup with distance attenuation model for audio channels where background talk was either audible or inaudible. The system had avatars to represent users, as well as hand tracking that animated and moved through the space as user performed gestures or spoke. Kim et al. [12] trained a machine learning model with synthetic data consisting of randomly generated 2000 faces to map human expressions to virtual avatars using a two layer, feed forward neural network. The trained model was then integrated with the system and verbal communication was integrated using the high level API (HLAPI) multiplayer library for Unity. The avatars were animated using head IK for head tracking and the Unity mecanim animation system for body tracking to trigger actions like talking and walking. Chang et al. [6] proposed design guidelines for cues when recognizing the availability of other users using eye gaze and body movements and capturing the attention of others using hand gestures, facial expressions, and proximity adjustment. A virtual meeting environment was developed by Kurzweg et al. [14] featuring a simulated avatar. A set of behaviors commonly observed in social meetings, such as attentiveness, arm crossing, and nervousness, was selected. Arrow and text symbols were created to direct participants' attention to a specific avatar.

Multisensory and Accessible Cues: Jung et al. [10] proposed designs for non-verbal cues eye contact, head nod, head shake, smile, and frown in iterations and proposed haptic and auditory cues for conveying the selected cues. Lee et al. [15] developed an attention guidance system for social VR with a light manipulator module to control environment lighting when user acknowledged the new speaker, point light if new speaker was completely out of view of the user, and spotlight when the target was within user's viewport. The system also had a spatial audio control module that used signaling sound to notify users of a participant awaiting their turn to speak, and to control the volume of the current speaker when other cues were being played. Wieland et al. [30] provided guidelines for inclusive cue design. They proposed auditory and haptic cues to support non-verbal communication and guiding attention to users such as with head and hand tracking for head nods and gestures, delivering haptic feedback, and auditory cues to alert users of another user's actions. Sasikumar et al. [23] developed a system with three components for remote collaboration, physiological sensing, and the virtual space with virtual objects. They represented physiological cues using three visualizations, a dynamic animated heart icon, a brain representation, and a moving circle to illustrate attention levels. They used animated avatars to deliver these cues by integrating multiple physiological sensors within the system, namely Shimmer3 for measuring galvanic skin response (GSR) and heart

rate, and Looxid EEG device for capturing attention. The collected data from Shimmer3 sensor via Bluetooth was streamed to Unity project using Lab Streaming Layer (LSL) and Looxid Link Unity Plugin was used to collect EEG data.

Spatial Awareness and Environmental Cues: Qin et al. [21] proposed extending the safety wall feature of VR systems to detect intruders. When a user was playing a VR game and an intruder entered the activity area, a safety boundary wall with a gap appeared in front of the user. The safety wall rotated in response to the user's movement, positioning the gap behind the intruder. Subsequently, a section of the safety boundary wall collapsed beneath the intruder's feet, alerting the user to the intruder's presence. Muller et al. [18] developed a mixed reality environment with virtual elements and placed physical objects as spatial cues in the physical space. They designed a memory game with cubes that had to be uncovered by touching them and finding matching cubes.

6.2 Evaluation

The evaluation of cue designs in multi-user VR and MR systems primarily focused on understanding how different cues impact user attention, social interaction, collaboration, and emotional experiences. Across the reviewed studies, evaluation approaches included both quantitative user studies and qualitative expert interviews and design workshops.

The most common evaluation method used in the included studies were user studies. Most studies conducted controlled user studies with participant numbers ranging from small pilot groups (6–8 participants) [35] to larger studies (over 90 participants) [8]. Tasks varied from joint attention exercises, collaborative problem solving, and social interaction in VR scenarios. Researchers used a combination of standardized scales such as NASA-TLX for cognitive workload [4, 21] and SUS for usability [33] along with eye tracking data [8, 27] and physiological data [23] from sensors for data collection during user studies.

In addition, expert reviews, semi-structured interviews, and design workshops were used to refine cue design and evaluate effectiveness of cues on user experience [6, 9, 30]. Some of the studies employed iterative design cycles, assessing cue effectiveness across multiple iterations to improve usability and alignment with natural social behaviors and usability of the system [10].

The cues were designed to support different aspects of user experience. Many studies evaluated whether users could effectively coordinate attention using visual and auditory cues, such as gaze-following, speech, or combined multisensory cues [4, 9, 27, 30, 33]. These assessments were also particularly focused on populations with social or cognitive differences such as participants with autism spectrum disorder (ASD) [5, 11, 25].

Cues targeting initiation of conversations [25], turn-taking [15], informal communication [8], and initiating conversations at a professional social VR events [6] were evaluated for their effectiveness to mimic social behavior in VR spaces. On the other hand, cues such as attention guidance, spatial awareness, and digital proxemics were evaluated for their impact on task efficiency and error reduction in collaborative tasks [31, 33].

7 Opportunities and Challenges

Cue-based activity recognition holds considerable promise for user activity recognition in multi-user VR and MR environments. At the same time, there are important design and usability challenges that must be addressed. We discuss core opportunities and challenges identified from the reviewed studies.

7.1 Opportunities

Improving Attention and Engagement: Strategically placing cues within immersive environments can improve users' attention to relevant events or partners, increasing engagement [27]. Visualizing cues that align with natural behavior such as gaze direction or pointing gestures supports users in following social and task-relevant signals even when those cues are not explicitly required by the task [5]. In multi-user settings, well-designed attention cues also support joint attention and coordination between participants [8, 11]. Embedding cues that convey social signals, such as gaze cones indicating the direction of another user's gaze, guides attention and reinforces both social presence and task coordination [4].

Supporting Accessibility and Inclusivity: Auditory and tactile cues show promise for increasing accessibility, particularly for people with visual impairments [10, 30] and young adults with ASD [5, 11, 25]. For example, auditory cues can help users detect non-verbal behaviors such as turn-taking or social initiation [10], while tactile feedback can signal touch interactions [30], expanding activity recognition beyond visual cues alone.

Communication and Co-Presence in Collaborative Tasks: Visualization of cues such as joint attention cues and annotation cues for remote users [33] can support shared understanding and communication between users [4]. These cues improve co-presence by making it easier to identify where others are looking, what they are focusing on, and how to coordinate actions in social and collaborative tasks. This reinforcement of co-presence serves to bridge the gap between virtual and physical social experiences. These cues can reinforce conversational flow [6], resolve ambiguity in referential communication [18], and help maintain situational awareness within groups [33].

Aligning Natural and Virtual Social Behaviors: Cue-based approaches also create opportunities to align system behavior with natural human movement and social patterns [8]. Participants in VR and MR contexts often exhibit behaviors that closely mirror real-world interaction, such as reaching out gestures [25], open body language during engagement [14], and gaze shifts to signal conversational turns [28]. Moreover, contextual cues such as shared spatial markers can help users identify target objects or navigate complex environments with greater confidence [35]. When cue design integrates in a way users naturally interact, the resulting experience feels more embodied and less intrusive [21].

7.2 Challenges

Design Concerns: One of the most prominent challenges in cue-based activity recognition systems is design complexity, encompassing both the selection of appropriate cues and their placement within immersive environments. Isolating the specific effects of individual cues versus the influence of system design or environmental context remains difficult [27]. Without clear experimental controls, it is challenging to determine whether observed user responses stem from the cue's informational content or simply from its spatial location [27]. Designing cues that are both informative and seamlessly integrated into the experience is thus a challenge. Closely related is the problem of cue overload, where multiple simultaneous signals can overwhelm users [10]. Users may struggle to remember or interpret concurrent visual, auditory, and tactile cues, especially in fast-moving or socially complex interactions and is therefore a key design concern [33].

Trade-offs in Multisensory Designs: Auditory cues can offer fast, attention-directing information, but they can also be intrusive in social contexts, particularly during conversation, when they may interrupt a speaker or compete with speech signals [15]. Likewise, spatial audio cues can be computationally expensive and difficult for users to distinguish from standard audio, reducing their practical effectiveness in resource-constrained systems [28]. Visual cues pose their own challenges. Rapid changes in visualization without explicit confirmation can confuse users [33], and lag in synchronized audio cues can disrupt the overall experience [4]. Additionally, physiological visualizations such as those designed to represent user affect are not always used frequently or interpreted

reliably by users, raising questions about their utility in broad application contexts [9, 23].

Technical Constraints: Effective cue-based activity recognition often depends on the availability of accurate tracking and sensing data, yet current hardware and algorithmic limitations hinder this goal. Full-body tracking and precise mapping of physical space remain technically demanding, especially in multi-user settings where occlusion, tracking drift, and environmental variability can degrade performance [35]. Similarly, machine learning algorithms used to classify behaviors may require extensive data and computation, limiting real-time responsiveness [25]. Head-mounted displays (HMDs) often occlude critical facial features, complicating efforts to localize expressions and integrate them into recognition systems [12].

Social Contexts and Privacy: Some users prefer implicit non-verbal cues over more explicit signals like directional gaze indicators [5]. Others, such as children with ASD, may miss socially driven cues altogether due to differences in eye contact or attentional patterns [11]. These variations challenge the development of universally effective cue designs and highlight the need for adaptable or customizable systems. In social contexts, cues intended to communicate internal states such as boredom or unavailability may feel intrusive or awkward if they explicitly highlight another user's affective condition [14]. Privacy concerns also arise in relation to haptic feedback and proximity cues [30], particularly in MR environments where personal space expectations intersect with shared physical space.

8 Discussion

The reviewed studies highlight several key considerations in the design, integration, and use of cues for activity recognition in multi-user VR and MR environments. While there is no standard cue-based approach for user activity recognition [27], the findings indicate that the effectiveness of cues is highly context-dependent, shaped by the task, user characteristics, interaction modality, and environmental constraints.

For example, visual gaze indicators or spatial overlays proved helpful in collaborative manipulation tasks allowing users to infer partners' attention or intentions [33], whereas in conversational settings, auditory cues or non-verbal signals were more effective [10]. This indicates the need for combining cues, rather than relying on a single channel to support natural and inclusive interactions.

Cues were consistently shown to act as both social enablers and trust mechanisms [35]. Users reported that cues providing information about collaborators' focus or movements increased confidence in shared spaces and promoted coordination [35]. Yet, the synthesis also suggests that cues do not always replace traditional interaction methods [4].

In some cases, verbal communication remained primary, with non-verbal or spatial cues acting as complementary channels to reinforce understanding. This is particularly evident in studies where language-first strategies were augmented with gaze or gesture cues [9], indicating that multisensory designs support richer interaction without causing a high cognitive load.

A recurrent theme in the results is the importance of intuitive and inclusive design. Studies involving young adults with ASD emphasized that cues must be explicit, interpretable, and consistent to be effective [5]. Similarly, cues that use familiar metaphors, such as safety boundaries or behavioral analogs from the real-world, were more easily understood and naturally integrated into user actions [21].

The results also revealed that human-agent interactions, while valuable, do not fully capture the subtlety of human-to-human behavior. For example, avatars conveying gestures or gaze may transmit basic information but often miss the nuanced social signals that occur in face-to-face interaction [11].

The studies further demonstrate that cue effectiveness is sensitive to cognitive factors and nuances of social interactions. In longer conversations or extended collaborative tasks, persistent or intrusive cues can be perceived as distracting [15]. Similarly, participants were more likely to disengage if the interaction steps required to interpret or respond to a cue became cumbersome [8]. These observations are further supported by results showing dynamic cues which appear or move in sync with the user's focus were noticed more reliably than static or subtle signals [23].

Environmental and contextual factors also played a significant role in the effectiveness of cues. The review revealed that lighting conditions and multisensory design affect cue perception for users with visual impairments [30]. HMD users, for example, were highly sensitive to personal space and field-of-view constraints [31]. Our results highlight that even when interaction is limited, such as when participants passively observed recorded meetings, cues can improve understanding of social dynamics and attention allocation [14], suggesting their utility beyond direct interaction tasks.

Another important insight from the results concerns the relationship between cues and cognitive workload. For example, the presence of additional virtual objects or dynamic visual overlays also supported spatial memory [18], indicating that task design and cue design are related to each other. Designers should therefore consider how cues contribute to overall cognitive load and ensure that any enhancements in awareness do not inadvertently increase user effort.

9 Conclusion

This scoping review synthesized research on cue-based approaches for user activity recognition in multi-user VR and MR systems. The most common types of cues developed for user activity recognition are visual, auditory, and tactile cues integrated in multi-user VR and MR systems as standalone cues or combined in multisensory designs. The cues developed are gaze for visual attention cues, full body movement and gestures for non-verbal cues, multisensory and accessible cues, and spatial awareness and environmental cues. These cues have been evaluated primarily through user studies to assess their effectiveness to mimic natural social behavior in VR and MR environments, and collaboration and coordination in multi-user contexts.

The main opportunities for cue-based approaches for user activity recognition are in their application for improving attention and engagement with other users and objects in the shared space, supporting accessibility and inclusivity for diverse user needs, improving communication and co-presence in collaborative tasks, and aligning virtual interactions more closely with natural social behaviors. However, some design challenges and technical constraints exist for realizing cue-based approaches such as trade-offs in multisensory designs and accommodating diverse user and social interaction contexts.

An emerging research area human-to-agent interactions represents a new paradigm for social and multi-user scenarios. Investigating how cues can convey intentions, attention, and affect between humans and embodied agents will be critical for designing collaborative systems in immersive environments. Exploring adaptive, multisensory, and context-sensitive cue designs in this emerging area will help bridge the gap between human-human and human-agent social dynamics, enabling more natural multi-user experiences.

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